# ECG record export

Sep 23, 2026 at 1:37 AM Apple Watch

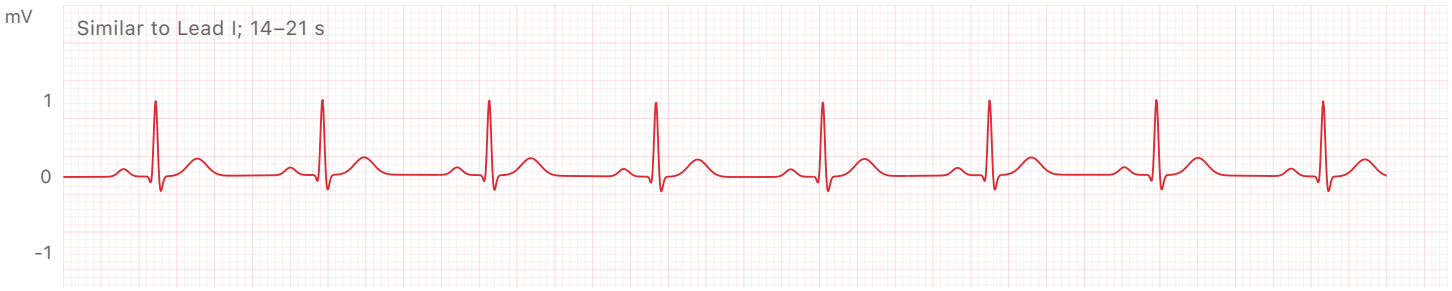
| Heart rate | Recording duration | Demo classification |
| --- | --- | --- |
| 68 bpm | 30 s | Sinus rhythm |

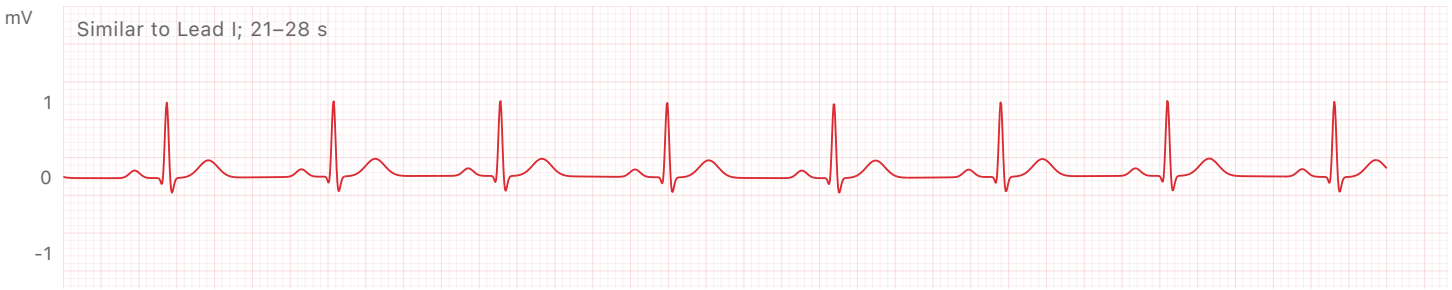
25 mm/s 10 mm/mV Print at actual size (100%)

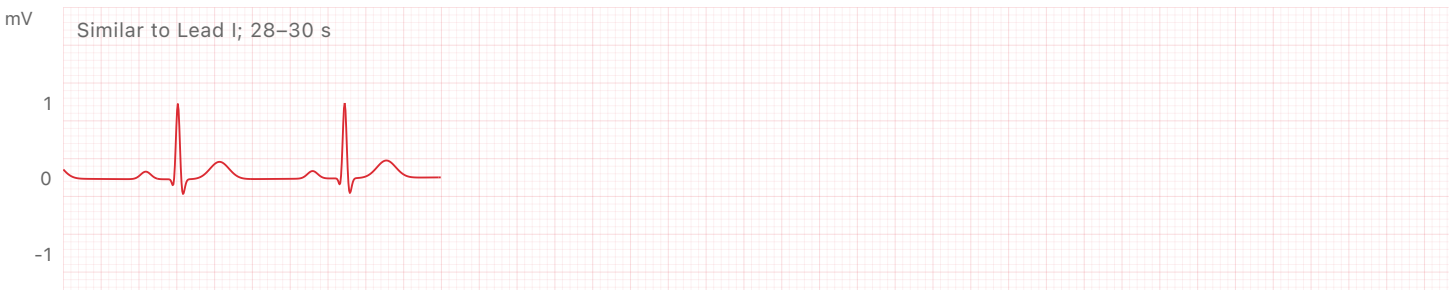









For reviewing records only. Not medical advice or a substitute for professional medical care.

# Analysis summary

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| Average heart rate in this record | Signal coverage | Analyzed duration |
|---|---|---|
| 68 bpm | 100% | 30 s |

| Metric | Result |
|---|---|
| Data sources | Apple Watch / Synthetic example |
| R peaks identified | 34 |
| Median RR | 883 ms |
| RR standard deviation | 1 ms |
| RMS of successive RR differences | 2 ms |

RR statistics do not exclude non-sinus beats and are not normal-beat HRV measures.

## HRV from this ECG

| SDNN | RMSSD | SD1 / SD2 |
|---|---|---|
| 1 ms | 2 ms | 1 / 1 ms |

27 seconds of clear signal, 31 beat intervals

This HRV estimate is from a short recording. Compare recordings of the same length taken under similar conditions.

## Premature-beat candidates and rhythm

| Rhythm reference score | Premature-beat candidates |
|---|---|
| 100 / 100 | Beats: 0 |

Detected beats: 34; included in analysis: 32.
Flagged beats: 0. Each beat is counted once.

Automatic flags require review against the original ECG. The rhythm score does not assess heart health.

## ECG segments

Range is the minimum to maximum in this recording, not a clinical reference interval. The central 80% is P10–P90.
— Unable to measure

## P wave

| Measurement | Median | Range | P10–P90 | Valid / total |
|---|---|---|---|---|
| P-wave duration | 121 ms | 117–121 ms | 117–121 ms | 34 / 34 |
| P-wave amplitude | 0.10 mV | 0.10–0.10 mV | 0.10–0.10 mV | 34 / 34 |

Adult reference: P wave < 120 ms
Reference values are general information. This measurement is not used to infer disease causes or offer medication or treatment advice. Single-lead estimates may contain errors and cannot assess health.

# ECG segments

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## PR interval

| Measurement | Median | Range | P10–P90 | Valid / total |
|---|---|---|---|---|
| PR interval | 188 ms | 188–191 ms | 188–188 ms | 34 / 34 |

Adult reference: PR 120–200 ms
Reference values are general information. This measurement is not used to infer disease causes or offer medication or treatment advice. Single-lead estimates may contain errors and cannot assess health.

## PR segment

| Measurement | Median | Range | P10–P90 | Valid / total |
|---|---|---|---|---|
| PR-segment duration | 66 ms | 66–70 ms | 66–70 ms | 34 / 34 |
| PR-segment level | 0.00 mV | 0.00–0.00 mV | 0.00–0.00 mV | 34 / 34 |

No universal normal range applies to this single-lead measurement
Reference values are general information. This measurement is not used to infer disease causes or offer medication or treatment advice. Single-lead estimates may contain errors and cannot assess health.

## QRS complex

| Measurement | Median | Range | P10–P90 | Valid / total |
|---|---|---|---|---|
| QRS duration | 98 ms | 94–98 ms | 94–98 ms | 34 / 34 |
| Q amplitude (estimated) | -0.07 mV | -0.07–-0.06 mV | -0.07–-0.06 mV | 34 / 34 |
| Q duration (estimated) | 20 ms | 16–20 ms | 17–20 ms | 34 / 34 |
| Main R-peak amplitude | 0.94 mV | 0.92–0.94 mV | 0.93–0.94 mV | 34 / 34 |
| QRS start to R peak | 43 ms | 39–43 ms | 39–43 ms | 34 / 34 |
| S amplitude (estimated) | -0.18 mV | -0.18–-0.18 mV | -0.18–-0.18 mV | 34 / 34 |
| S duration (estimated) | 31 ms | 31–35 ms | 31–35 ms | 34 / 34 |

Adult reference: QRS < 120 ms
Reference values are general information. This measurement is not used to infer disease causes or offer medication or treatment advice. Single-lead estimates may contain errors and cannot assess health.

# ECG segments

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## ST segment / J point

| Measurement | Median | Range | P10–P90 | Valid / total |
|---|---|---|---|---|
| ST-segment duration | 59 ms | 55–59 ms | 55–59 ms | 34 / 34 |
| J-point level | -0.01 mV | -0.01–-0.00 mV | -0.01–-0.00 mV | 34 / 34 |
| Level 60 ms after J | — | — | — | 0 / 34 |
| Level 80 ms after J | — | — | — | 0 / 34 |
| ST voltage change rate | — | — | — | 0 / 34 |

Reasons for missing values: Occurrences: 34. The T-wave start is unclear, so the ST measurement window cannot be determined.; Occurrences: 34. This position is too close to the T wave to measure ST separately.

No universal normal range applies to this single-lead measurement
Reference values are general information. This measurement is not used to infer disease causes or offer medication or treatment advice. Single-lead estimates may contain errors and cannot assess health.

## T wave

| Measurement | Median | Range | P10–P90 | Valid / total |
|---|---|---|---|---|
| T-wave duration | 219 ms | 219–223 ms | 219–219 ms | 34 / 34 |
| T-wave amplitude | 0.23 mV | 0.23–0.23 mV | 0.23–0.23 mV | 34 / 34 |
| T-wave duration ratio | 1.00 × | 0.97–1.00 × | 1.00–1.00 × | 34 / 34 |

No universal normal range applies to this single-lead measurement
Reference values are general information. This measurement is not used to infer disease causes or offer medication or treatment advice. Single-lead estimates may contain errors and cannot assess health.

## QT interval

| Measurement | Median | Range | P10–P90 | Valid / total |
|---|---|---|---|---|
| QT interval | 371 ms | 371–375 ms | 371–375 ms | 34 / 34 |

No universal normal range applies to this single-lead measurement
Reference values are general information. This measurement is not used to infer disease causes or offer medication or treatment advice. Single-lead estimates may contain errors and cannot assess health.

# ECG segments

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## QTc (corrected QT)

| Measurement | Median | Range | P10–P90 | Valid / total |
|---|---|---|---|---|
| QTc (Fridericia) | 387 ms | 387–391 ms | 387–391 ms | 31 / 34 |
| QTc (Bazett) | 395 ms | 395–399 ms | 395–399 ms | 31 / 34 |

Reasons for missing values: Occurrences: 2. The QT end is unclear, or nearby RR intervals do not meet the correction conditions. See Measurement method for details.; Occurrences: 1. This beat's QT or nearby continuous, stable RR intervals are missing, so correction is unavailable.

Adult QTc: men about 350–450 ms; women about 360–460 ms
Reference values are general information. This measurement is not used to infer disease causes or offer medication or treatment advice. Single-lead estimates may contain errors and cannot assess health.

## TP segment

| Measurement | Median | Range | P10–P90 | Valid / total |
|---|---|---|---|---|
| TP-segment duration | 324 ms | 320–324 ms | 320–324 ms | 33 / 34 |

Reasons for missing values: Occurrences: 1. This T-wave end or the next P-wave start is unclear, or there may be a signal gap between them.

No universal normal range applies to this single-lead measurement
Reference values are general information. This measurement is not used to infer disease causes or offer medication or treatment advice. Single-lead estimates may contain errors and cannot assess health.

## RR / PP intervals

| Measurement | Median | Range | P10–P90 | Valid / total |
|---|---|---|---|---|
| RR interval | 883 ms | 879–883 ms | 880–883 ms | 33 / 34 |
| PP interval | 883 ms | 879–883 ms | 880–883 ms | 33 / 34 |

Reasons for missing values: Occurrences: 1. The previous beat is missing, the P wave is unclear, or the signal has a gap between beats.

Adult resting reference: RR 600–1000 ms (60–100 bpm)
Reference values are general information. This measurement is not used to infer disease causes or offer medication or treatment advice. Single-lead estimates may contain errors and cannot assess health.

# Measurement method

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These values are estimated automatically by the app. Accuracy has not been validated on real Apple Watch data, and the values cannot determine whether an ECG is normal.

| Metric | Result |
| --- | --- |
| Data sources | Apple Watch |
| Sampling rate | 256 Hz |
| Analysis version | 0.6.0-research |

## Detection and scoring rules

Rhythm reference score = 100 × (1 – flagged beats ÷ reviewable beats), rounded. Flags include premature-beat candidates, marked interval changes or local waveform changes. The pause after a candidate is not counted again. This describes the share of beats not flagged by these rules in this recording. It is not a heart-health score, and 100 does not mean the ECG is normal.

At least 20 seconds of clear signal, 80% signal coverage, 15 reviewable beats and 8 stable reference beats are required. At least 80% of non-edge beats must be reviewable. Poor signal does not reduce the score; missing PR or QT values and higher or lower HRV do not add or subtract points.

Compare with nearby beats: the beat must arrive over 15% and at least 80 ms early, with a longer following interval or a wider, changed QRS. Wide candidates require a local width of at least 120 ms, over 20% wider than the reference, and template correlation below 0.90. Narrow candidates require a width under 120 ms, correlation of at least 0.90 and a longer following interval. These research rules cannot by themselves confirm atrial or ventricular premature beats.

Rule version: rhythm-review-1

## P wave

The start, end and peak of the P wave are estimated. If it is too small, its boundaries are unclear or it overlaps nearby waves, the affected values are left blank.

## PR interval

Uses the P-wave start and QRS start from the same beat. If either is unclear, no PR interval is calculated. PR includes the P wave itself and the following PR segment.

## PR segment

Measured from the P-wave end to QRS start. The level is the mean voltage of this segment relative to the local TP baseline. Positive values are above the baseline; negative values are below it.

## QRS complex

QRS start and end are estimated first. Opposite-direction waves before and after the main peak are candidates for Q and S. Values retain their original polarity. If the main peak points down, Q, R and S labels need manual review. A single lead cannot assess the electrical axis or R-wave progression across chest leads.

# Measurement method

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## ST segment / J point

The QRS end estimates the J point. Voltage is measured 60 and 80 ms later. The change rate comes from the voltage difference between these positions: positive means upward and negative downward. Positions too close to the T wave are not used for ST measurements.

## T wave

The T-wave start, peak and end are found separately. The duration ratio is start-to-peak time divided by peak-to-end time. It is near 1 when the two parts have similar lengths. Multiple peaks require checking the original trace.

## QT interval

Each beat is measured separately, from its QRS start to its T-wave end, before results are summarized. If the T-wave end is unclear, QT is not calculated.

## QTc (corrected QT)

Uses QT from the same beat and the immediately preceding RR, both in seconds: Fridericia = QT / RR^(1/3); Bazett = QT / √RR. The algorithm requires continuous neighboring RR intervals, changes no greater than 10%, and intervals of 500–1500 ms. These are algorithm conditions, not a normal range.

## TP segment

Measured from this T-wave end to the next P-wave start. No value is produced if either boundary is unclear, the signal has a gap, or the interval is shorter than 40 ms.

## RR / PP intervals

RR and PP are calculated from their own consecutive peaks. They do not substitute for one another, and intervals are not joined across signal gaps.

# Recording notes

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## Sep 23, 2026 at 1:37 AM

Duration: 2 min
Caffeine

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